

Blockchain based Electronic Voting System Design with Smart Contracts

Wan Auzan Bin W Abdulah
School of Electrical Engineering
College of Engineering
Universiti Teknologi MARA
40450 Shah Alam, Selangor
wanauzanwabdulah@gmail.com

Syed Farid Syed Adnan
School of Electrical Engineering
College of Engineering
Universiti Teknologi MARA
40450 Shah Alam, Selangor
syed_farid@uitm.edu.my

Abstract— The voting process serves as the mechanism to translate public opinion into actionable measures, significantly impacting democracy. Ensuring the transparency, reliability, and security of the voting procedure is essential to maintain the integrity and legitimacy of the entire system. Blockchain-based voting systems have gained popularity due to the immutability and advancements in blockchain technology. Creating a secure electronic voting system that offers the transparency and flexibility of electronic systems while preserving fairness and privacy has long been a challenge. This paper introduces a method for collecting data from various sources and databases using transparent Blockchain technology. The proposed Blockchain-based solution is dependable, secure, and ensures privacy. The objective of this paper is to leverage blockchain technology to construct a transparent and secure voting system. The development, testing, and deployment of smart contracts are executed using the Truffle framework, while a local blockchain network is established using the Ganache tool. MetaMask is employed for account validation. This paper also presents the cost analysis of the voting transactions.

Keywords—Blockchain, Voting, Ganache, Truffle, Smart Contracts, MetaMask

I. INTRODUCTION

Democratic elections are one of the major sectors that blockchain is expected to revolutionise in the years to come, according to researchers who claim that this technology is the future of technology [1]. Blockchain means a decentralised distributed ledger system that is controlled in a decentralized way and detects blockchain integrity that can improve data security, openness, and transparency. The underlying blockchain technology is a cryptocurrency built from the architectural design of a distributed database that records transactions and stores them in blocks [2]. The interest in blockchain as a technology increased tremendously after the original Blockchain paper was published in the year 2008 by Satoshi Nakamoto [3]. Electronic voting, in addition to reducing human error, enables fast data analysis. Many electronic voting systems now use a centralised electronic voting system for data processing; however, centralised servers are vulnerable to malicious assaults and DDOS attacks, thus distributed architecture should be employed instead to limit the danger of shutdown after an attack [4]. This is where blockchain technology takes part in creating a completely decentralized and open, online voting system based on blockchain. Due to its intrinsic capabilities of maintaining anonymity and keeping a fragmented and widely scattered track of data across all nodes, blockchain might play a crucial role in electronic voting. As a result, blockchain technology is particularly efficient at dealing with the

possibility of multiple uses of a vote token and attempts to alter the outcome's transparency [5].

This paper's main goals are to shine a light on blockchain technology, explain its use in the storage and secure transmission of votes, and conceive a blockchain-based voting architecture that offers immutability, auditability, trust, and transparency.

Elections using a paper ballot system are harder to conduct in countries with larger populations. The paper ballot system is time-consuming and unreliable as there is a chance that it might be rigged by vote tampering, booth capturing, etc [6]. Although each region, city, or district can run its own node of the system to distribute the load, the decentralisation of the Blockchain network will allow real-time viewing of election results across the entire nation [7]. Additionally, there are a few drawbacks to digital voting. Some negative techniques used throughout the voting process include fake voting, cost reductions, producing quicker results, etc. In fact, several attacks may severely impact voting or voting calculations by causing damage to smart or IoT (Internet of Things) systems to benefit themselves [8]. Another one of the most critical issues with the voting system continues to be low voter turnout. The event's limited duration and location, as well as mistrust of the voting process, are the causes of the poor voter turnout. Remote electronic voting has the best chance of eliminating these factors, but it cannot be guaranteed to be completely secure, impervious to hacking, to ensure voting confidentiality and transparency for society, or to ensure the system's overall fault tolerance.

The proposed solution is to use blockchain technology to create a secure and transparent system that gives different stakeholders visibility into the entire process of recording and storing votes [1]. The government sector can benefit from the present Industrial Revolution 4.0 development, particularly in terms of how it shows election results. The sophistication of blockchain technology will enhance security system improvements and voting procedures [2]. As voters can cast their votes at any time and from any location using blockchain-based voting, it may result in a higher voter turnout than the existing method [9].

There is a lack of transparency and integrity in the current system of voting. The main purpose of this study is to propose an Ethereum blockchain technology using smart contract to provide fairness besides anonymity, integrity, security, and privacy.

This study also aims to design a computerized voting system to speed up the process of vote counting and to develop a remote electronic voting system to avoid low voter turnout.

II. LITERATURE REVIEW

The Blockchain-based electronic voting application is decentralized, meaning that no data is stored on a single server. Candidate and voter registration must be done in advance. Account creation requires the verification of identity. The authorized individual is then required to authenticate eligible users by validating a coin or token after confirming their identity. Blockchain will make sure that using two coins or tokens throughout the verification procedure is not permitted. A user is only permitted to cast one vote [10]. The blockchain setup addresses the problem of vote rigging [9].

Most of the research papers used Ethereum as a platform that allows programmers to create decentralized applications on the Ethereum platform [11]. Most approaches of research papers that have utilised smart contracts such as [3], [12], [1], [9] and [8]. The self-executing software, or smart contracts, is created inside the blockchains. These are used for code of conduct agreements between two parties and are similar to conventional business contracts. When the predetermined criteria are satisfied, the smart contracts immediately go into action. Without the need for a centralized authority, smart contracts enable the trusted execution of agreements and transactions between unreliable or unknown parties [11].

For the registration process, several research papers used tokens or coins after the registration process. Authors in [2], [6], [11] used coin or card-shaped tokens which is quite useful to avoid any double vote in the process of voting. After identity verification, the authorized person should authenticate eligible users by providing a coin or token. Using this coin or token each user can vote only once. Blockchain's verification process will ensure that double spending on this token is not possible.

There are different ways for each research paper for the verification phase. For example, E. Febriyanto et al. [2] proposed an index fingerprint as verification before voting. Then, the voters will get a card-shaped token to vote, that is only once. Other works by T. Luo [13] proposed a proxy multi-signature method to reduce the load on the system and enhance the capacity of democratic decision-making. The scheme is based on the Schnorr signature, and the proxy signature is delegated by certificate to allow interoperability with current blockchain systems. Meanwhile, J. Abegunde et al. [14] used OTP to send to voters' known phones to verify to voters. Other than these research papers, most of them used the conventional method which is to go through identification authentication with the Certifying Authority, which has previously acquired the user's identity data from the government.

Lastly, S. T. Alvi et al. [15] proposed sidechain technology. The authors proposed three chains involved: the Registration Chain, the Voting Chain, and the Vote Management Chain. The Registration Chain and the Voting Chain serve as Sidechains, and the Vote Management Chain serves as the Main Chain. This sidechain technology reduces the cost consumption and extends the capabilities of the main chain.

Therefore, from most of the research, the elements of the Blockchain voting process include the process of voter registration, voter authenticity and the voting process itself.

III. PROPOSED METHOD

This work proposed a voting system using the Ethereum blockchain network. It is an open-source, decentralized blockchain network that supports smart contracts. Ether is the digital currency here on the Ethereum platform (ETH). It uses Proof of Work (PoW) as the consensus algorithm, allowing anyone with enough computing power to solve a problem rapidly to add a new block to the network.

A. Flowchart

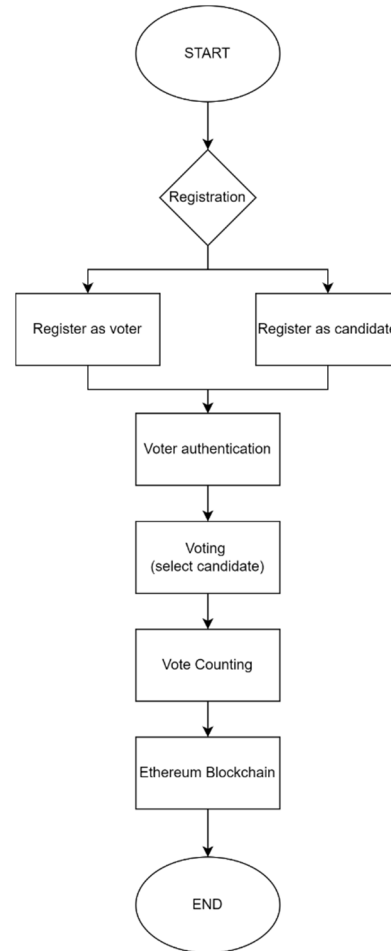


Fig 1. Flowchart of the voting system based on blockchain.

Fig. 1 illustrates the Ethereum blockchain-based voting system's flow process, spanning from registration to vote counting. The initial stage consists of two primary flows: voter registration and candidate registration. A voter, defined as a legally registered citizen, is identified by name on the voter list.

B. Voter Registration

Before voting, each eligible voter is provided with a unique public address and a private key. These credentials are crucial for subsequent authentication. Sufficient ethers are allocated in each voter's account, allowing for a single transaction. It is imperative to maintain the confidentiality of

both the private key and public address, preserving the anonymity of the voters.

Within this blockchain, a smart contract, referred to as the "registration contract," is operational. During the registration process, this contract stores comprehensive voter data, serving as the foundation for voter authentication during the balloting phase. Additionally, the registration contract maintains an inventory of candidate data within the blockchain. This information becomes accessible when voters cast their ballots.

Notably, candidates also undergo the same registration process as voters. In the blockchain's genesis block, critical candidate information, including candidate number, party symbol, and public key, is securely stored.

C. Voter Authentication

The voter authentication phase took place where the role of the smart contract ran in the blockchain. Using a computer or other internet-connected device, the voter logs into the voting system using a private key. The information about the legitimate voters in the Genesis block is verified using smart contracts incorporated into the blockchain. The voter is given a list of candidates if the information is accurate.

D. Voting Phase

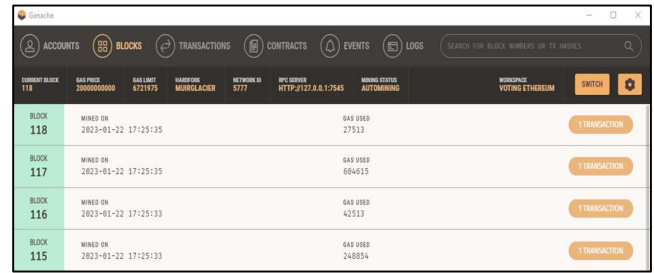
After completing the authentication procedure through the registration contract, the voting phase is utilized to record the encrypted votes of the electorate. The voter selects a candidate from the list of candidates and then casts their ballot. Lastly, only when the transaction ID is confirmed by the miner (anyone) and recognized by all the miners is the block transaction successful.

E. Blockchain Implementation

This proposed work is implemented using the Ethereum blockchain network. It is an open-source, decentralized blockchain network that supports smart contracts. In Ether, a cryptocurrency is being used here on the Ethereum platform (ETH). Solidity-based smart contracts for Ethereum are used to cast ballots. This implementation tests the smart contracts before deploying them on the blockchain using the truffle framework. Decentralized application development, testing, and deployment are made easier by the truffle framework. It offers a setting for blockchain network growth. Building smart contracts, compiling built-in contracts, linking those contracts, and deploying them can all be done using the truffle development framework.

The environment of the truffle includes Ganache, a personal blockchain for the advancement of Ethereum. An Ethereum client can be visualized as a client. Ganache is used to test the truffle-based decentralized application. Then, decentralized applications are created to deploy contracts. The application is then launched on an Ethereum client like Geth after being tested on Ganache. For testing, Ganache was used as both physical and virtual blockchain. In this study, ten external user accounts are created and available. A distinct Ethereum address and private key have been assigned to every account in Ganache. Every account has 100 "fake" ethers already loaded.

IV. RESULTS & DISCUSSION



Block	Transaction	Gas Used
118	0x6f4b14a3f1103a40e23b3c9c9b2266771d1d6bb0797b725ce619d15f71bdaa2a	27513
117	0xd22f75e96167081c63ef938fe7c835c6a999a0c8	684615
116	0xb8447fa532b95b4402caf9490d0ab6876ecf9837	42513
115	0x120923	248854

Fig 2. Deployed smart contract in Ganache

Fig. 2 shows the deployment of smart contracts in the blockchain. When a smart contract is deployed in Ganache, it is added to the local blockchain in four different blocks:

1. The deployment transaction, which delivers the smart contract bytecode to the blockchain and generates a new contract address, is contained in the first block.
2. The second block is the one that mines the contract address as part of the deployment transaction and adds it to its list of transactions.
3. The third block adds the contract address to its list of transactions and confirms the deployment operation.
4. The fourth block includes the contract address in its list of transactions and confirms the deployment operation.

This procedure makes sure the smart contract is deployed successfully and is functional. Additionally, it helps developers test their smart contracts in a virtual environment that strongly matches the primary Ethereum network.

```

3_selection.js
=====
Deploying 'Election'
> transaction hash: 0x6f4b14a3f1103a40e23b3c9c9b2266771d1d6bb0797b725ce619d15f71bdaa2a
> Blocks: 0
> contract address: 0xd22f75e96167081c63ef938fe7c835c6a999a0c8
> block number: 117
> block timestamp: 1674379535
> account: 0xb8447fa532b95b4402caf9490d0ab6876ecf9837
> balance: 99.5315547
> gas used: 604615 (0x939c7)
> gas price: 20 gwei
> value sent: 0 ETH
> total cost: 0.0120923 ETH

> Saving migration to chain.
> Saving artifacts
=====
> Total cost: 0.0120923 ETH

Summary
=====
> Total deployments: 2
> Final cost: 0.01706938 ETH

```

Fig. 3. Migration using Truffle

Fig. 3 shows a certain amount of ether spent from the account which is used to deploy the smart contract in the blockchain network for each migration. After a smart contract is activated, the blockchain is updated via migrations. We must construct a migration, which is a numbered Javascript file, for each smart contract. These files are automatically called in order by the truffle framework. Migration has features for deploying smart contracts to a given network and for tracking the deployed contracts' most recent versions. Additionally, it enables developers to maintain the data and state of earlier versions of smart contracts while deploying new versions of the contracts on the blockchain.

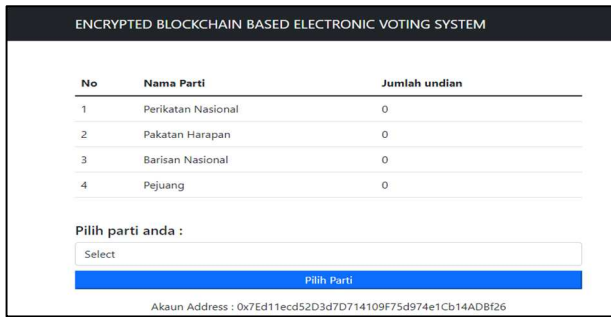


Fig. 4. The user interface of the proposed voting system

Fig. 4 represents the user interface (UI) for the proposed system of voting for the users to interact with to cast their votes. Every vote is recorded as a transaction on the blockchain. Each transaction contains the voter's account address, which is used to verify the voter's identity, and the vote itself, which is typically encoded as a set of data that represents the voter's choice.

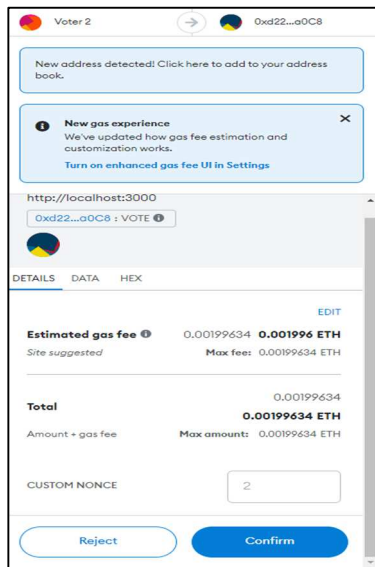


Fig. 5. Metamask transaction confirmation

Fig. 5 shows the transaction that the user needs to be confirmed to cast a vote using the account that has been addressed and given. The transaction is then sent to the Ethereum blockchain network, where it is broadcast to all the nodes on the network. The nodes on the network validate the transaction by checking that the voter's account has sufficient funds (usually Ether) to pay for the gas fee associated with the transaction and that the vote is valid according to the rules of the voting smart contract. Once the transaction is validated, it is added to a block and mined by a miner. Once the block is mined, the voting smart contract will execute the vote and the vote will be recorded on the blockchain.

A. Transparency, Accuracy and Integrity

In order to evaluate the performance of the voting blockchain system concerning transparency, accuracy, and integrity, we need to show that votes on the user interface are tallied with the transaction in the Ganache platform.

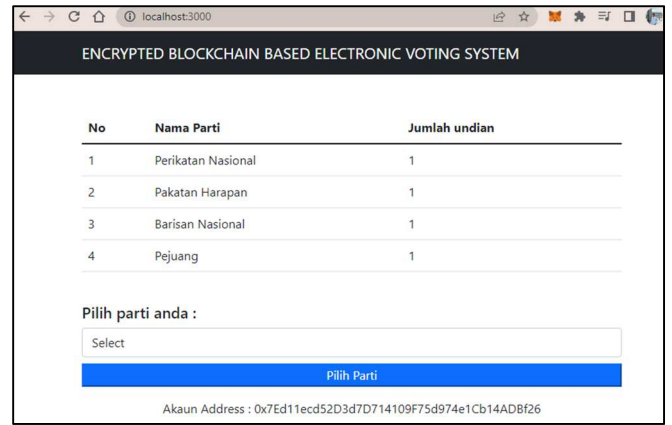


Fig. 6. The result of the first four votes

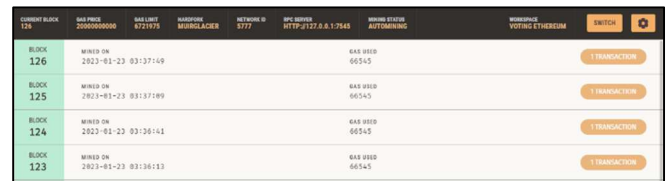


Fig. 7. The log of the first four votes on Ganache

At the end of the initial run of the simulation, four voters voted for the four candidates to ensure that it corresponded with the votes cast at the user interface. On the user interface of the webpage, there will be 4 candidates to vote for which are Perikatan Nasional, Pakatan Harapan, Barisan Nasional and Pejuang. After each voter votes for each candidate, the result will be displayed on the Ganache platform. In Fig. 7, the Ganache platform showed the transaction logs for four blocks with a different hash transaction to show the four votes have been verified and put into the block. Therefore, the votes in the user interface are tallied with the transaction in the Ganache platform.

Merkle tree is implemented using blockchain technologies to guarantee data integrity. The Merkle tree concept was developed for Ethereum to offer quick and reliable proof that a transaction will take place in a node. All the transactions in a block are hashed and included in the Merkle tree. A Markle tree must be created in the transaction if a node needs to determine whether a transaction has been updated or not. This makes it very simple to validate or invalidate a vote. In our system, the Markle tree is employed to maintain the integrity of the election process.

B. Anonymity and Privacy

Voter entry into the blockchain in the proposed system is done anonymously because the hash function can provide anonymity. In the blockchain network, the public key is given to every voter for the MetaMask or during the registration procedure to ensure only authorized individuals can vote, but it is not linked to their personal information, and the voter's anonymity is protected. The encryption ensures that only authorized parties can access the transaction data and the voter's identity cannot be traced back to the vote. This ensures that the voting process is secure and transparent while maintaining the anonymity of the voter.

Fig. 8 exhibits the voter interface and the voting accounts created for users, respectively, are used to evaluate the qualities of anonymity and coercion-resistance. These

provide zero information regarding the identity of the voter. In a similar vein, as can be seen in the transaction log in Fig. 8, there was no voter-identifying information in the vote data that the smart contract entered the ledger. Combining these elements shows that the voter's identification information is replaced by the Ganache account in a planned manner and is unrelated to the vote.

The proposed approach gives voters anonymity and privacy. In the voting method, votes are entered anonymously onto the blockchain. In the blockchain network, the voter's public key acts as their identity and the hash value they produce during registration acts as their data. The voter's privacy and anonymity are protected in this way.

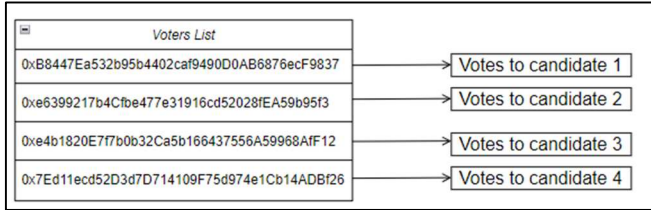


Fig. 8: Anonymity of voter

C. Cost Analysis

The currency used on the Ethereum network is called ether (ETH). While the blockchain and the EVM's computation are costed in ETH, the execution charge is calculated in terms of gas. Gas and ETH are purposefully kept apart so that the price of ETH might change depending on conditions in the global market and so that the cost of gas is directly correlated to the cost of computation. In practice, one unit of gas equates to the execution of one computational phase.

We observed variations in cost in Ethereum while increasing the number of candidates and changing the smart contract coding. As the number of candidates increases, the size of the smart contract increases, and more computational effort is required to execute it. This means that more gas is required to deploy the smart contract. A general positive trend of a very slight increase in gas cost was observed with an increasing candidate count.

In Table I, we show the gas costs and the corresponding prices in RM for the deployment of the smart contract.

TABLE I ACTUAL COST FOR CONTRACT DEPLOYMENT

Smart Contract	Reason	Number of candidates	Cost (Ether)	Cost (RM)
1	Use less loop and use uint8	2	0.01391544	96.98
2	Use uint256 and use loop	2	0.01520766	105.99
3	Use uint256 and more candidates	5	0.01799484	125.41

Optimizing the code of a smart contract is an important aspect of developing decentralized applications (dApps) on the Ethereum blockchain. Gas cost is one of the key factors to consider when deploying a smart contract, as it can affect the feasibility and scalability of the contract. By reducing the gas cost, dApps can be more cost-effective and accessible to users.

For the first smart contract, the smart contract coding uses a smaller number of functions and variables to reduce the

complexity of the smart contract's code which will result in a reduced cost of ether. Each operation in the smart contract code requires a certain amount of gas to execute. By reducing the number of operations, the gas cost can be reduced. This can be achieved by removing unnecessary operations, using more efficient algorithms, and simplifying the logic of the smart contract.

Table I shows the second smart contract uses more gas than the first one as the second one uses more complex coding. When the smart contract code is complex, it requires more computational effort to execute it, and therefore, more gas is required. This is because a large number of operations in the smart contract code can increase the gas cost, as each operation requires a certain amount of gas to execute. Smart contracts with complex logic can require more gas to execute, as it takes more effort for the Ethereum Virtual Machine (EVM) to process.

The third smart contract used more gas than the two smart contracts before as the last one has more candidates and uses complex coding. Increasing the number of candidates in a voting blockchain system can increase the gas used when deploying the smart contract. As the number of candidates increases, the size of the smart contract increases, and more computational effort is required to execute it. This means that more gas is required to deploy the smart contract. If the smart contract has to store the name, address and other information of each candidate, this will increase the storage required, and as a result, more gas will be required.

Therefore, using less complex coding can help to reduce the gas cost of deploying and executing a smart contract on the Ethereum blockchain. Reducing the gas cost is an important aspect of developing decentralized applications (dApps) on the Ethereum blockchain, as it can affect the feasibility and scalability of the contract.

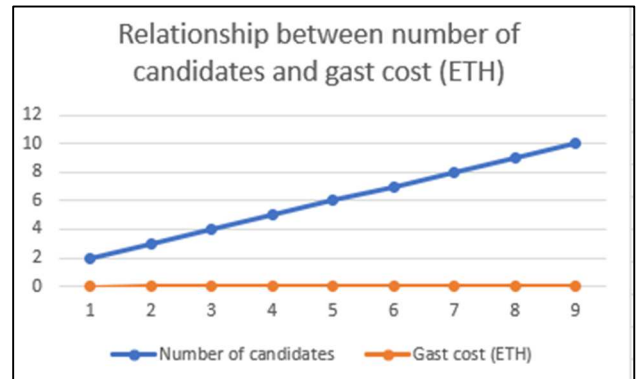


Fig. 9: Line chart number of candidates and gas cost (ETH)

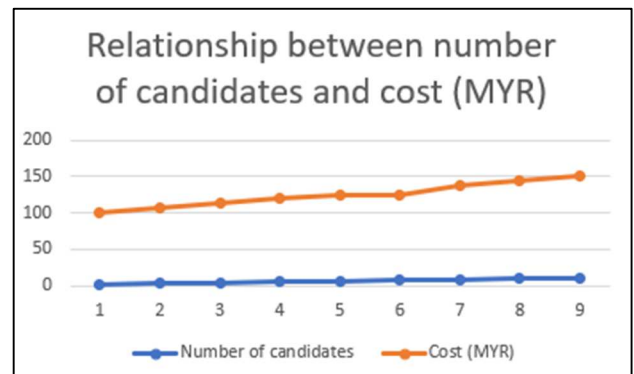


Fig. 10: Line chart number of candidates and gas cost (MYR)

Fig. 9 and 10 shows the line chart to display the relationship between the number of candidates and gas cost. The x-axis represents the number of candidates, and the y-axis represents the gas cost. Based on the figure above, when the number of candidates increases, the gas cost also increases. This can be seen by a steep line that goes up the chart. This is because the amount of computational work required to process the transaction increases as the number of candidates increases. Thus, this has led to an increase in gas costs as more computational resources is required to execute the transactions.

V. CONCLUSION AND FUTURE WORK

This paper presents a blockchain-based voting system that runs on Ethereum. It demonstrates how centralized voting system limitations can be addressed by blockchain technology. Ganache is used to implement this voting mechanism and test its functionality. Votes are stored as immutable and tamper-proof after being cast by the voters. The security concerns with the present EVM technology are resolved by this. Additionally, it promptly counts the votes, cutting down on the length of the election process.

Gas consumption in a smart contract can be considerably decreased by optimizing the code. This can be done by employing a variety of strategies, including fewer loops, effective data structures, and fewer state-changing operations. When creating smart contracts, it is necessary to bear in mind that gas costs are an important factor to take into account because they have a direct impact on how much it costs to complete transactions on the blockchain. To reduce the amount of gas used, it is crucial to properly design and optimize the code.

In the future, the frontend user interface (UI) of the application can be enhanced to display election data and any additional authentication techniques can be included to further boost voting system security and credibility.

ACKNOWLEDGEMENT

The authors would like to express their thanks and appreciation to Ministry of Higher Education (MOHE) for providing the grant FRGS/1/2019/ICT04/UITM/02/7 and UiTM Shah Alam for supporting this research work.

REFERENCES

- [1] Agbesi Samuel and Asante George, "Electronic Voting Recording System Based on Blockchain Technology," in 2019 12th CMI Conference on Cybersecurity and Privacy (CMI): Aalborg University Copenhagen, Copenhagen, Denmark, 28-29 November 2019., 2019.
- [2] E. Febriyanto, Triyono, N. Rahayu, K. Pangaribuan, and P. A. Sunarya, "Using Blockchain Data Security Management for E-Voting Systems," in 2020 8th International Conference on Cyber and IT Service Management, CITSM 2020, Oct. 2020. doi: 10.1109/CITSM50537.2020.9268847.
- [3] W. Zhang et al., "A Privacy-Preserving Voting Protocol on Blockchain," in IEEE International Conference on Cloud Computing, CLOUD, Sep. 2018, vol. 2018-July, pp. 401–408. doi: 10.1109/CLOUD.2018.00057.
- [4] H. te Wu and C. Y. Yang, "A blockchain-based network security mechanism for voting systems," in Proceedings - 2018 1st International Cognitive Cities Conference, IC3 2018, Dec. 2018, pp. 227–230. doi: 10.1109/IC3.2018.00-15.
- [5] M. Ramalingam, D. Saranya, and R. Shankararam, "An Efficient and Effective Blockchain-based Data Aggregation for Voting System," in 2021 International Conference on System, Computation, Automation and Networking, ICSCAN 2021, Jul. 2021. doi: 10.1109/ICSCAN53069.2021.9526450.
- [6] G. S. Tomar and Institute of Electrical and Electronics Engineers, "Decentralized Voting Application Using Blockchain," in 2021 10th IEEE International Conference on Communication Systems and Network Technologies (CSNT) proceedings, 2021.
- [7] V. Sliusar, A. Fyodorov, A. Volkov, P. Fyodorov, and V. Pascari, "Blockchain Technology Application for Electronic Voting Systems," in Proceedings of the 2021 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering, ElConRus 2021, Jan. 2021, pp. 2257–2261. doi: 10.1109/ElConRus51938.2021.9396400.
- [8] S. T. Alvi, M. N. Uddin, and L. Islam, "Digital voting: A blockchain-based E-voting system using biohash and smart contract," in Proceedings of the 3rd International Conference on Smart Systems and Inventive Technology, ICSSIT 2020, Aug. 2020, pp. 228–233. doi: 10.1109/ICSSIT48917.2020.9214250.
- [9] T. Vairam, S. Sarathambekai, and R. Balaji, "Blockchain based Voting system in Local Network," in 2021 7th International Conference on Advanced Computing and Communication Systems, ICACCS 2021, Mar. 2021, pp. 363–366. doi: 10.1109/ICACCS51430.2021.9441912.
- [10] A. M. Al-Madani, A. T. Gaikwad, V. Mahale, and Z. A. T. Ahmed, "Decentralized E-voting system based on Smart Contract by using Blockchain Technology," in Proceedings of the 2020 International Conference on Smart Innovations in Design, Environment, Management, Planning and Computing, ICSIDEMPC 2020, Oct. 2020, pp. 176–180. doi: 10.1109/ICSIDEMPC49020.2020.9299581.
- [11] K. Patidar and S. Jain, "Decentralized E-Voting Portal Using Blockchain; Decentralized E-Voting Portal Using Blockchain," 2019.
- [12] S. T. Alvi, M. N. Uddin, L. Islam, and S. Ahamed, "A Privacy-Aware Digital Voting System employing Blockchain and Smart Contracts," in 2020 IEEE Asia-Pacific Conference on Computer Science and Data Engineering, CSDE 2020, Dec. 2020. doi: 10.1109/CSDE50874.2020.9411634.
- [13] T. Luo, "An Efficient Blockchain Based Electronic Voting System Using Proxy Multi-signature," in 2021 3rd International Academic Exchange Conference on Science and Technology Innovation, IAECS 2021, 2021, pp. 513–516. doi: 10.1109/IAECST54258.2021.9695917.
- [14] J. Abegunde, J. Spring, and H. Xiao, "SEVA: A Smart Electronic Voting Application Using Blockchain Technology," in Proceedings - 2021 IEEE International Conference on Blockchain, Blockchain 2021, 2021, pp. 353–360. doi: 10.1109/Blockchain53845.2021.00056.
- [15] S. T. Alvi, M. N. Uddin, L. Islam, and S. Ahamed, "A blockchain based cost effective digital voting system using SideChain and smart contracts," in Proceedings of 2020 11th International Conference on Electrical and Computer Engineering, ICECE 2020, Dec. 2020, pp. 467–470. doi: 10.1109/ICECE51571.2020.9393081.